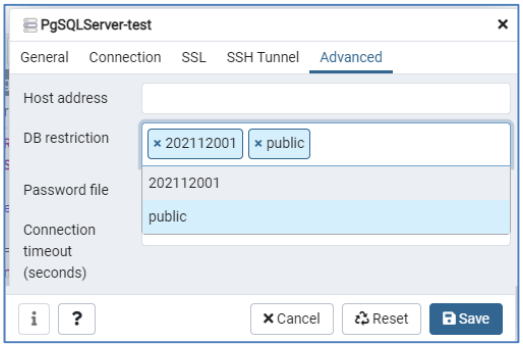
Lab03 | Querying operations on relations .1

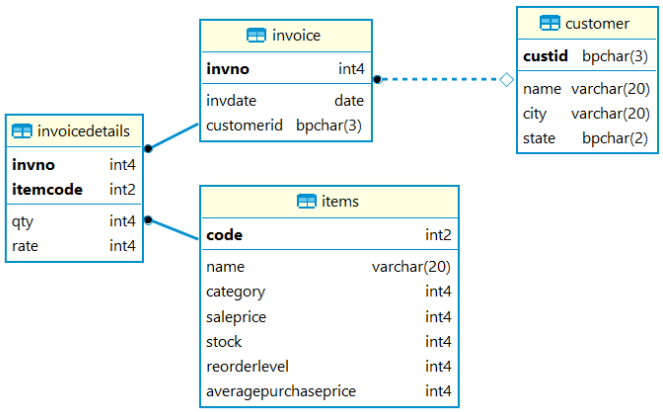
DS604 Introduction to Data Management, Autumn'2025; pm_jat @ daiict

In this lab, you learn and practice querying databases using Relational Algebra and SQL. For practice purposes, you are given access to a database named as **`public`** (on the server 10.100.71.21). Where various schemas are populated with some sample data that you can use for practice purposes.



Before accessing, add this database to DB Restriction as following: while you are disconnected from the server, right click on server-name; click on properties; go to advanced tab; add public database; typically, as shown below and save it.

Here is schema diagram for Sales database for your reference



Your goal here is to write relational algebra expressions and SQL statements for answering the following queries. [Algebraic expressions can be handwritten and scanned for submission]

1. Give the details of items having the price > 1000
2. Give the details of items having the price > 1000 and belonging to category 5;
3. List the Invoices (number, date) of a customer id = 'C05'
4. List all items (code, qty, rate) for invoice number = 1
5. List the Invoices (number, date) of the customer named Harsh.
6. List all invoices (inv no, date, customer rid) that have item code = 1103
7. List all items (inv no, item code, qty) that customer 'C05' has bought
8. List all customers (customer id, name, city) that have bought item 1101
9. List all items (item code, item name) sold on 2011-08-23
10. List all items (item code, item name) sold in month '2011-07' from category=3
11. List customers (customer id, name, city) who are from Gujarat and have bought item items from category=3
12. List the items (code, name) of items purchased by a customer named 'Dev' on the date 2011-08-23

[Submission Required: Query Expressions in Relational Algebra and SQL]

Submission Instructions: Submit everything in a single PDF file.